

Jeroen Monnee

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ORCID | [INSPIRE-HEP](#)

I work at the interface of string theory, quantum gravity, and algebraic geometry. My research develops Hodge-theoretic and geometric methods for studying infinite-distance limits, emergent strings, flux compactifications, and quantum consistency conditions in Calabi–Yau compactifications.

Appointments and Education

Academic appointments

2024–present ♦ **Postdoctoral Researcher**, University of Hamburg, Germany
Host: Prof. Timo Weigand

Education

- 2020–2024 ♦ **PhD in Theoretical Physics**, Utrecht University, the Netherlands
Supervisor: Prof. Thomas Grimm
🔗 Thesis: *Asymptotic Hodge Theory in String Compactifications and Integrable Systems*
Degree awarded: August 2024
- 2018–2020 ♦ **MSc in Theoretical Physics**, Utrecht University, the Netherlands
Supervisor: Prof. Thomas Grimm
🔗 Thesis: *The Distribution of Vacua in Asymptotic Flux Compactifications of Type IIB/F-Theory*
- 2015–2018 ♦ **BSc in Physics and BSc in Mathematics**, Utrecht University, the Netherlands

Research Publications

- 1 L. Kaufmann, **J. Monnee**, T. Weigand, and M. Wiesner, “Quantum obstructions for $N = 1$ infinite distance limits – Part II: Kähler obstructions,” *Phys. Rev. D*, vol. 113, no. 12, p. 126 028, 2026. 🔗 DOI: 10.1103/ypyx-mg6r. arXiv: 2603.13470 [hep-th].
- 2 L. Kaufmann, **J. Monnee**, T. Weigand, and M. Wiesner, “Quantum obstructions for $N = 1$ infinite distance limits – Part I: g_s obstructions,” Mar. 2026. arXiv: 2603.12315 [hep-th].
- 3 **J. Monnee**, T. Weigand, and M. Wiesner, “K points and type IIB/heterotic duality with NS5-branes,” *Phys. Rev. D*, vol. 113, no. 6, p. 066 002, 2026. 🔗 DOI: 10.1103/kcbz-jf16. arXiv: 2510.02435 [hep-th].
- 4 **J. Monnee**, T. Weigand, and M. Wiesner, “Physics and geometry of complex structure limits in type IIB Calabi-Yau compactifications,” *JHEP*, vol. 03, p. 063, 2026. 🔗 DOI: 10.1007/JHEP03(2026)063. arXiv: 2509.07056 [hep-th].
- 5 B. Hassfeld, **J. Monnee**, T. Weigand, and M. Wiesner, “Emergent strings in Type IIB Calabi-Yau compactifications,” *JHEP*, vol. 01, p. 140, 2026. 🔗 DOI: 10.1007/JHEP01(2026)140. arXiv: 2504.01066 [hep-th].
- 6 T. W. Grimm and **J. Monnee**, “Finiteness theorems and counting conjectures for the flux landscape,” *JHEP*, vol. 08, p. 039, 2024. 🔗 DOI: 10.1007/JHEP08(2024)039. arXiv: 2311.09295 [hep-th].
- 7 T. W. Grimm and **J. Monnee**, “Bi-Yang-Baxter models and $Sl(2)$ -orbits,” *JHEP*, vol. 11, p. 123, 2023. 🔗 DOI: 10.1007/JHEP11(2023)123. arXiv: 2212.03893 [hep-th].
- 8 T. W. Grimm and **J. Monnee**, “Deformed WZW models and Hodge theory. Part I,” *JHEP*, vol. 05, p. 103, 2022. 🔗 DOI: 10.1007/JHEP05(2022)103. arXiv: 2112.00031 [hep-th].
- 9 T. W. Grimm, **J. Monnee**, and D. van de Heisteeg, “Bulk reconstruction in moduli space holography,” *JHEP*, vol. 05, p. 010, 2022. 🔗 DOI: 10.1007/JHEP05(2022)010. arXiv: 2103.12746 [hep-th].

Academic Experience

Invited talks and seminars

- Jun. 2026 ♦ **Utrecht University** — String Theory Seminar
Emergent strings and quantum obstructions in infinite-distance limits of Type IIB string theory
- Mar. 2026 ♦ **IFT Madrid** — SPS Journal Club
Quantum obstructions for $\mathcal{N} = 1$ infinite-distance limits
- Feb. 2026 ♦ **KU Leuven** — Joint Belgian HEP–TH Seminar
Emergent strings and KK towers in Type II/III/IV limits of Type IIB string theory
- Nov. 2025 ♦ **Harvard University** — Swampland Group Lunch
Towers of light states in Type IIB complex-structure degenerations
- Oct. 2025 ♦ **Seminar Series on String Phenomenology** — online seminar
The II, III, IV of Calabi–Yau degenerations  [video](#)
- Mar. 2025 ♦ **Heidelberg University** — String Theory and Beyond the Standard Model Seminar
Asymptotics in complex-structure moduli space and degenerations of Calabi–Yau manifolds
- Nov. 2024 ♦ **University of Hamburg** — Swampland Seminar
Charting the string landscape with asymptotic Hodge theory
- May 2024 ♦ **Swamplandia 2024**, Bavaria — invited conference talk
Complexity of the flux landscape
- Dec. 2023 ♦ **Seminar Series on String Phenomenology** — online seminar
Complexity of the flux landscape  [video](#)
- Jun. 2022 ♦ **Seminar Series on String Phenomenology** — online seminar
Bulk reconstruction in moduli-space holography  [video](#)

Contributed talks

- Jul. 2026 ♦ **String Phenomenology 2026**, Paris — parallel-session talk
EFT strings and quantum obstructions to 4D $\mathcal{N} = 1$ limits
- Jul. 2025 ♦ **String Phenomenology 2025**, Boston — parallel-session talk
Emergent strings in Type II_b limits of Type IIB string theory
- Jul. 2023 ♦ **String Phenomenology 2023**, Daejeon — parallel-session talk
Finiteness of flux vacua and the tadpole conjecture
- Jul. 2022 ♦ **String Phenomenology 2022**, Liverpool — parallel-session talk
Hodge theory and deformed WZW models

Selected poster presentations

- May 2026 ♦ **Strings and Geometry 2026**, Uppsala
Quantum Kähler obstructions to $\mathcal{N} = 1$ infinite-distance limits
- May 2024 ♦ **Strings and Geometry 2024**, Hamburg
Complexity of the flux landscape
- May 2022 ♦ **Strings and Geometry 2022**, Utrecht — gong show and poster presentation
Asymptotic Hodge theory and deformed WZW models

Academic Experience (continued)

Teaching experience

Lecturer and tutor

- 2025–2026 ◇ **University of Hamburg**
Student Seminar: Supersymmetry and Superstrings — MSc Physics
Course instruction and supervision of student presentations.
- 2025 ◇ **University of Hamburg**
Mathematical Methods B — BSc Nanophysics
Lecturer and tutor.

Teaching assistant

- 2026 ◇ **University of Hamburg**
Theoretical Physics II: Quantum Mechanics — BSc Physics
- 2020–2023 ◇ **Utrecht University**
String Theory, General Relativity, Quantum Field Theory, and Field Theory in Particle Physics
— MSc Theoretical Physics

Thesis supervision

Day-to-day supervisor of four MSc and two BSc theses at Utrecht University.

MSc theses

- 2023–2024 ◇ Lars Verboven — *Complexity in F-theory flux compactifications and counting vacua*
- ◇ Luuk Lagendijk — *On the connection between Hodge Theory and integrable systems*
- 2021–2022 ◇ Christoph Sieling — *Chiral matter in F-theory compactifications*
- 2020–2021 ◇ Arno Hoefnagels — *Conformal quantum mechanics in holography and Hodge theory*

BSc theses

- 2024 ◇ Lara Timmers — *Einstein's general relativity: symmetries in curved spacetime*
- 2021 ◇ Hanneke Schrotten — *Geodesics in semi-Riemannian geometry and links to general relativity*

Advanced schools

- 2023 ◇ **Amsterdam String Cosmology school 2023** — *The Quantum Gravity Swampland and its consequences for the observable world*
- 2022, 2023 ◇ **DRSTP Postgraduate School in Theoretical High-Energy Physics**, the Netherlands
- 2020 ◇ **Amsterdam-Brussels-Geneva-Paris Doctoral School** — *Quantum Field Theory, Strings and Gravity*

Academic service and organization

- 2025–present ◇ **String Theory Seminar**, University of Hamburg and DESY
Co-organizer of the weekly seminar series.
- 2021–2023 ◇ **Alumni Talks**, Utrecht University
Co-organizer of a recurring seminar series in which alumni of the theoretical-physics MSc and PhD programmes presented their career paths beyond academia, providing career orientation for current students.
- 2022 ◇ **Strings and Geometry 2022**, Utrecht University
Co-organizer of the workshop.

Additional Information

Languages and technical skills

Languages ◇ Dutch (native); English (fluent); Spanish (basic).

Tools ◇ Mathematica; Python (working knowledge); LaTeX.

Outreach and public engagement

2016–present ◇ **Vierkant voor Wiskunde**

Volunteer organizer and instructor at an annual mathematics summer camp for children aged 8–12.

References

◇ **Prof. Thomas W. Grimm**

Supervisor during bachelor thesis, master thesis and PhD

Professor at Utrecht University, Chair ‘Quantum Gravity and Geometry’.

Institute for Theoretical Physics, Utrecht University.

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